Thirth Patel

Software Engineer | AI / ML | Backend & Full-Stack | B.Tech ISE (AI & Robotics)
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SUMMARY

AI/ML and full-stack software engineer (B.Tech ISE, CGPA 8.82) who ships tested, documented, deployed systems. Team Lead at Smart India Hackathon 2026: built a consent-first government data-interoperability gateway with 187 REST APIs, LLM + RAG legal search and 768 automated tests in CI. Hands-on with Python, FastAPI, PostgreSQL, Docker, React and LLM agents; 2 industry internships. Seeking SDE / AI-ML internships.

EDUCATION

Presidency University, Bengaluru 2024 - 2028 (expected)
B.Tech, Information Science & Engineering (AI & Robotics) **CGPA: 8.82 / 10**
Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Web Technologies, Artificial Intelligence

EXPERIENCE

Full-Stack Web Development Intern | Technofenatics Solutions LLP (Sattix), Bengaluru Jul 2026 - Sep 2026

- Shipped features to Shorthand Pro, a live production PHP + SQLite web app, through a ticket-based Agile workflow: made community links admin-configurable via the existing key-value settings store (no code deploy and no database migration needed for future changes).
- Resolved 2 role-based dashboard UI defects using Chrome DevTools; root-caused one to browser/server caching and fixed it with filemtime() cache busting, now the reusable pattern for CSS releases.
- Rated "working - continuing" at the mid-internship review (7 of 13 interns) for daily stand-ups and on-time delivery.

Web Development Intern | Amdox Technologies, Remote Jan 2026 - Feb 2026

- Built JobConnect, a full-stack job portal (Node.js, Express, REST APIs, JWT + bcrypt authentication, role-based access control) with multi-filter job search and a 5-stage hiring pipeline (Applied to Hired).

PROJECTS

CivicLens | *Government data-interoperability gateway* | *Smart India Hackathon 2026, Team Lead* github.com/thirthpatel2-web/Civiclens
Python, FastAPI, PostgreSQL, pgvector, Redis, NiceGUI, LLM (Groq), Whisper, Docker, GitHub Actions

- Led the team and architected a gateway where departments reuse verified citizen data only after consent: identity resolution, consent ledger, data-quality rules, schema mapping and per-hop request tracing; delivered 31/34 SIH26129 spec capabilities, 31/31 E2E checks passing.
- Built 187 REST API operations + WebSockets, 50 web screens, voice input (Whisper STT) in 10 Indian languages and a RAG legal analyzer (pgvector semantic search) over 38,238 Supreme Court records that refuses uncited answers.
- Enforced quality with 768 automated tests (unit, integration, end-to-end), ruff, mypy and a GitHub Actions CI/CD pipeline; deployed with Docker Compose + Caddy HTTPS; made the API ~10x faster under load (13 ms/request) by caching a per-request Argon2 hash.

Cascade Failure Env | *AI-agent benchmark* | *Meta x Hugging Face OpenEnv Hackathon* huggingface.co/spaces/ThirthPatel/cascade-failure-env
Python, OpenEnv, LLM agents, Reinforcement Learning, Docker, Hugging Face Spaces

- Designed a reinforcement-learning environment where LLM agents act as on-call SREs in a 12-service microservice dependency graph; deterministic graders, shaped rewards, 3 difficulty tiers; deployed on HF Spaces.

IRIS - Disaster Intelligence System | *FusionX Hackathon 2026, AI & ML track* github.com/thirthpatel2-web/disaster-intel
Python, FastAPI, WebSockets, Leaflet, Chart.js, LLM (Groq)

- Real-time multi-hazard risk scoring (5 hazards, 14 Indian cities) with 72-hour early warnings, evacuation and resource planning, 13 REST endpoints, a WebSocket alert stream and an LLM analyst with rule-based fallback.

RoadScan AI | *Computer vision* github.com/thirthpatel2-web/Road_scan_AI-
Python, OpenCV, Flask, TensorFlow / Keras (CNN)

- Pothole detection and road-condition grading from one image in ~1 s: OpenCV thresholding, morphology and contour filtering to reject shadows and lane marks; auto-switches to a Keras CNN when a trained model exists.

Signal & EthicaMind | *Applied NLP and responsible AI* github.com/thirthpatel2-web
Python, scikit-learn, Flask, React, Gemini API

- Signal: NLP resume-to-job matching across 25 roles (TF-IDF + cosine similarity) with skill-gap analysis. EthicaMind: responsible-AI chat with a crisis-safety layer before any LLM call and multi-model failover.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Java, C++, PHP, HTML/CSS
AI / ML: Machine Learning, Deep Learning, NLP, Computer Vision, LLMs, RAG, Vector Search (pgvector), Embeddings, AI Agents, Reinforcement Learning, scikit-learn, TensorFlow/Keras, OpenCV, Whisper, Ollama, Groq, Gemini
Backend, Web & Data: FastAPI, Flask, Node.js, Express, REST APIs, WebSockets, PostgreSQL, Redis, MongoDB, SQLite, Alembic, JWT, React, Vite, NiceGUI, Leaflet, Chart.js, three.js
DevOps & Quality: Git, GitHub Actions (CI/CD), Docker, Docker Compose, Caddy, Linux, pytest, Playwright, ruff, mypy
Concepts: Data Structures & Algorithms, OOP, DBMS, OS, Computer Networks, System Design, Distributed Systems, Microservices, API Design, Authentication & RBAC, Caching, Testing, Agile/Scrum

ACHIEVEMENTS & CERTIFICATIONS

- Smart India Hackathon 2026: Team Lead; Top 142 teams at Presidency University's internal round (CivicLens, SIH26129).
- Hackathons: Meta x Hugging Face OpenEnv (Cascade Failure Env), FusionX 2026 (IRIS), Hack Devengers 2.0 (1,500+ entries).
- Certifications: Infosys Springboard - Database Fundamentals: Getting Started with SQL, Introduction to MySQL; Corizo - Web Development internship (2026).